

Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)

SHAH

SY-131

SY-131

SY01311

OH

Design: SY-131

Standard Survey Report

18 June, 2019

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-131
Project:	SHAH	TVD Reference:	RTKB @ 483.7ft
Site:	SY-131	MD Reference:	RTKB @ 483.7ft
Well:	SY-131	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	SY-131	Database:	EDMPRODR5K

Project	SHAH, SHAH		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	Nahrwan - U. Arab Emer.		
Map Zone:	Zone 39N (48 E to 54 E)		

Site		SY-131, Shah-E			
Site Position:		Northing:	2,535,699.77 m	Latitude:	22° 54' 23.344 N
From:	Map	Easting:	787,820.78 m	Longitude:	53° 48' 20.607 E
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in	Grid Convergence:	1.09 °

Well	SY-131					
Well Position	+N/-S	0.0 ft	Northing:	2,535,699.77 m	Latitude:	22° 54' 23.344 N
	+E/-W	0.0 ft	Easting:	787,820.78 m	Longitude:	53° 48' 20.607 E
Position Uncertainty		0.0 ft	Wellhead Elevation:	ft	Ground Level:	453.2 ft

Wellbore	OH				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	WMM_2005	31/12/2009	1.56	34.99	42,625.70906600

Design	SY-131				
Audit Notes:					
Version:	Phase:	ACTUAL	Tie On Depth:	0.0	
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)	
	0.0	0.0	0.0	225.48	

Survey Program	Date	18/06/2019			
From (ft)	To (ft)	Survey (Wellbore)	Tool Name	Description	
100.0	4,756.5	Survey GYRO (OH)	CT_GYRO_MS	Continuous Gyro Multishot	
4,840.0	7,324.9	Survey MWD (OH)	MWD ISCWSA	MWD - Standard	

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00	
100.0	0.12	245.27	100.0	0.0	-0.1	0.1	0.12	0.12	0.00	
125.0	0.13	241.52	125.0	-0.1	-0.1	0.2	0.05	0.04	-15.00	
150.0	0.19	236.85	150.0	-0.1	-0.2	0.2	0.25	0.24	-18.68	
175.0	0.30	224.92	175.0	-0.2	-0.3	0.3	0.48	0.44	-47.72	
190.0	0.33	222.29	190.0	-0.2	-0.3	0.4	0.22	0.20	-17.57	
13-3/8" CONDUCTOR										
200.0	0.35	220.78	200.0	-0.3	-0.4	0.5	0.22	0.20	-15.05	
225.0	0.41	215.45	225.0	-0.4	-0.5	0.6	0.28	0.24	-21.32	

Survey Report

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Well:	SY-131	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	SY-131	Database:	EDMPRODR5K

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
250.0	0.48	200.39	250.0	-0.6	-0.6	0.8	0.54	0.28	-60.24	
275.0	0.40	179.99	275.0	-0.8	-0.6	1.0	0.70	-0.32	-81.60	
300.0	0.17	208.92	300.0	-0.9	-0.6	1.1	1.06	-0.92	115.72	
325.0	0.25	239.31	325.0	-0.9	-0.7	1.2	0.54	0.32	121.56	
350.0	0.34	222.59	350.0	-1.0	-0.8	1.3	0.49	0.36	-66.88	
375.0	0.40	198.85	375.0	-1.2	-0.9	1.4	0.65	0.24	-94.96	
400.0	0.22	177.74	400.0	-1.3	-0.9	1.5	0.84	-0.72	-84.44	
425.0	0.09	251.37	425.0	-1.3	-0.9	1.6	0.85	-0.52	294.52	
450.0	0.16	249.65	450.0	-1.4	-1.0	1.6	0.28	0.28	-6.88	
475.0	0.17	230.96	475.0	-1.4	-1.0	1.7	0.22	0.04	-74.76	
500.0	0.15	190.23	500.0	-1.5	-1.1	1.8	0.45	-0.08	-162.92	
525.0	0.12	168.31	525.0	-1.5	-1.1	1.8	0.24	-0.12	-87.68	
550.0	0.11	11.46	550.0	-1.5	-1.0	1.8	0.90	-0.04	-627.40	
575.0	0.14	283.90	575.0	-1.5	-1.1	1.8	0.70	0.12	-350.24	
600.0	0.13	215.67	600.0	-1.5	-1.1	1.9	0.61	-0.04	-272.92	
625.0	0.11	178.64	625.0	-1.6	-1.1	1.9	0.31	-0.08	-148.12	
650.0	0.09	90.03	650.0	-1.6	-1.1	1.9	0.56	-0.08	-354.44	
675.0	0.11	344.85	675.0	-1.6	-1.1	1.9	0.64	0.08	-420.72	
700.0	0.13	263.81	700.0	-1.5	-1.1	1.9	0.63	0.08	-324.16	
725.0	0.12	196.09	725.0	-1.6	-1.2	1.9	0.56	-0.04	-270.88	
750.0	0.13	114.34	750.0	-1.6	-1.2	1.9	0.66	0.04	-327.00	
775.0	0.08	22.44	775.0	-1.6	-1.1	1.9	0.62	-0.20	-367.60	
800.0	0.09	297.97	800.0	-1.6	-1.1	1.9	0.46	0.04	-337.88	
825.0	0.14	216.85	825.0	-1.6	-1.2	1.9	0.62	0.20	-324.48	
850.0	0.12	165.86	850.0	-1.6	-1.2	2.0	0.45	-0.08	-203.96	
875.0	0.16	108.58	875.0	-1.7	-1.1	2.0	0.55	0.16	-229.12	
900.0	0.08	13.64	900.0	-1.7	-1.1	1.9	0.74	-0.32	-379.76	
925.0	0.13	286.07	925.0	-1.6	-1.1	1.9	0.60	0.20	-350.28	
950.0	0.13	232.18	950.0	-1.6	-1.2	2.0	0.47	0.00	-215.56	
975.0	0.09	151.91	975.0	-1.7	-1.2	2.0	0.58	-0.16	-321.08	
1,000.0	0.10	82.58	1,000.0	-1.7	-1.2	2.0	0.43	0.04	-277.32	
1,025.0	0.13	20.76	1,025.0	-1.7	-1.1	2.0	0.48	0.12	-247.28	
1,050.0	0.11	319.52	1,050.0	-1.6	-1.1	1.9	0.49	-0.08	-244.96	
1,075.0	0.09	223.25	1,075.0	-1.6	-1.2	2.0	0.60	-0.08	-385.08	
1,100.0	0.09	159.04	1,100.0	-1.7	-1.2	2.0	0.38	0.00	-256.84	
1,125.0	0.15	114.18	1,125.0	-1.7	-1.1	2.0	0.43	0.24	-179.44	
1,150.0	0.11	53.08	1,150.0	-1.7	-1.1	1.9	0.55	-0.16	-244.40	
1,175.0	0.11	318.17	1,175.0	-1.6	-1.1	1.9	0.65	0.00	-379.64	
1,200.0	0.15	277.60	1,200.0	-1.6	-1.1	1.9	0.39	0.16	-162.28	
1,225.0	0.14	233.04	1,225.0	-1.6	-1.2	2.0	0.44	-0.04	-178.24	
1,250.0	0.11	193.49	1,250.0	-1.7	-1.2	2.0	0.36	-0.12	-158.20	
1,275.0	0.09	135.81	1,275.0	-1.7	-1.2	2.1	0.39	-0.08	-230.72	

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Well:	SY-131	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	SY-131	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
1,300.0	0.06	75.01	1,300.0	-1.7	-1.2	2.1	0.32	-0.12	-243.20
1,325.0	0.09	344.31	1,325.0	-1.7	-1.2	2.0	0.44	0.12	-362.80
1,350.0	0.11	291.54	1,350.0	-1.7	-1.2	2.0	0.36	0.08	-211.08
1,375.0	0.12	234.97	1,375.0	-1.7	-1.2	2.1	0.44	0.04	-226.28
1,400.0	0.13	198.45	1,400.0	-1.7	-1.3	2.1	0.32	0.04	-146.08
1,425.0	0.12	136.56	1,425.0	-1.8	-1.3	2.1	0.52	-0.04	-247.56
1,450.0	0.13	67.11	1,450.0	-1.8	-1.2	2.1	0.57	0.04	-277.80
1,475.0	0.10	20.41	1,475.0	-1.8	-1.2	2.1	0.38	-0.12	-186.80
1,500.0	0.10	300.24	1,500.0	-1.7	-1.2	2.1	0.52	0.00	-320.68
1,525.0	0.13	244.91	1,525.0	-1.7	-1.2	2.1	0.44	0.12	-221.32
1,550.0	0.11	194.71	1,550.0	-1.8	-1.3	2.1	0.41	-0.08	-200.80
1,575.0	0.09	159.46	1,575.0	-1.8	-1.3	2.2	0.25	-0.08	-141.00
1,600.0	0.15	84.79	1,600.0	-1.8	-1.2	2.1	0.61	0.24	-298.68
1,625.0	0.13	43.97	1,625.0	-1.8	-1.2	2.1	0.40	-0.08	-163.28
1,650.0	0.11	329.95	1,650.0	-1.7	-1.2	2.1	0.58	-0.08	-296.08
1,675.0	0.13	290.96	1,675.0	-1.7	-1.2	2.1	0.33	0.08	-155.96
1,700.0	0.13	231.71	1,700.0	-1.7	-1.3	2.1	0.51	0.00	-237.00
1,725.0	0.10	203.77	1,725.0	-1.8	-1.3	2.2	0.25	-0.12	-111.76
1,750.0	0.10	141.69	1,750.0	-1.8	-1.3	2.2	0.41	0.00	-248.32
1,775.0	0.11	96.94	1,775.0	-1.8	-1.2	2.2	0.32	0.04	-179.00
1,800.0	0.11	36.27	1,800.0	-1.8	-1.2	2.1	0.44	0.00	-242.68
1,825.0	0.13	349.46	1,825.0	-1.8	-1.2	2.1	0.39	0.08	-187.24
1,850.0	0.15	298.74	1,850.0	-1.7	-1.2	2.1	0.49	0.08	-202.88
1,875.0	0.10	245.28	1,875.0	-1.7	-1.3	2.1	0.48	-0.20	-213.84
1,900.0	0.12	174.46	1,900.0	-1.7	-1.3	2.1	0.51	0.08	-283.28
1,925.0	0.15	149.96	1,925.0	-1.8	-1.3	2.2	0.26	0.12	-98.00
1,950.0	0.16	131.90	1,950.0	-1.8	-1.2	2.2	0.20	0.04	-72.24
1,975.0	0.11	100.85	1,975.0	-1.9	-1.2	2.2	0.35	-0.20	-124.20
2,000.0	0.05	39.89	2,000.0	-1.9	-1.2	2.1	0.38	-0.24	-243.84
2,025.0	0.01	158.03	2,025.0	-1.9	-1.2	2.1	0.22	-0.16	472.56
2,050.0	0.13	211.37	2,050.0	-1.9	-1.2	2.2	0.50	0.48	213.36
2,075.0	0.27	194.81	2,075.0	-2.0	-1.2	2.2	0.60	0.56	-66.24
2,100.0	0.48	182.40	2,100.0	-2.1	-1.2	2.4	0.90	0.84	-49.64
2,125.0	0.65	168.69	2,125.0	-2.4	-1.2	2.5	0.86	0.68	-54.84
2,150.0	0.66	156.95	2,150.0	-2.6	-1.1	2.6	0.54	0.04	-46.96
2,175.0	0.62	158.22	2,175.0	-2.9	-1.0	2.8	0.17	-0.16	5.08
2,200.0	0.55	162.72	2,200.0	-3.1	-0.9	2.9	0.33	-0.28	18.00
2,225.0	0.74	171.68	2,225.0	-3.4	-0.9	3.0	0.86	0.76	35.84
2,250.0	1.08	182.52	2,250.0	-3.8	-0.8	3.3	1.52	1.36	43.36
2,275.0	1.46	183.52	2,275.0	-4.4	-0.9	3.7	1.52	1.52	4.00
2,300.0	1.61	181.73	2,300.0	-5.0	-0.9	4.2	0.63	0.60	-7.16
2,325.0	1.90	191.18	2,325.0	-5.8	-1.0	4.8	1.64	1.16	37.80
2,350.0	2.26	194.56	2,349.9	-6.7	-1.2	5.5	1.52	1.44	13.52

Survey Report

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Well:	SY-131	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	SY-131	Database:	EDMPRODR5K

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
2,375.0	2.76	199.83	2,374.9	-7.7	-1.5	6.5	2.20	2.00	21.08	
2,400.0	3.29	204.60	2,399.9	-8.9	-2.0	7.7	2.35	2.12	19.08	
2,425.0	3.80	208.91	2,424.8	-10.3	-2.7	9.2	2.30	2.04	17.24	
2,450.0	4.33	211.58	2,449.8	-11.8	-3.6	10.9	2.25	2.12	10.68	
2,475.0	4.95	212.17	2,474.7	-13.6	-4.7	12.9	2.49	2.48	2.36	
2,500.0	6.01	211.26	2,499.6	-15.6	-5.9	15.2	4.25	4.24	-3.64	
2,525.0	7.14	210.96	2,524.4	-18.0	-7.4	17.9	4.52	4.52	-1.20	
2,550.0	8.33	211.22	2,549.2	-20.9	-9.2	21.2	4.76	4.76	1.04	
2,575.0	9.44	211.73	2,573.9	-24.2	-11.2	25.0	4.45	4.44	2.04	
2,600.0	10.43	212.28	2,598.5	-27.9	-13.5	29.1	3.98	3.96	2.20	
2,625.0	11.39	213.01	2,623.1	-31.9	-16.0	33.8	3.88	3.84	2.92	
2,650.0	12.22	214.07	2,647.5	-36.1	-18.8	38.8	3.43	3.32	4.24	
2,675.0	12.70	214.98	2,671.9	-40.6	-21.9	44.1	2.07	1.92	3.64	
2,700.0	12.98	215.58	2,696.3	-45.1	-25.1	49.5	1.24	1.12	2.40	
2,725.0	13.34	216.14	2,720.7	-49.7	-28.4	55.1	1.53	1.44	2.24	
2,750.0	13.78	217.17	2,745.0	-54.4	-31.9	60.9	2.01	1.76	4.12	
2,775.0	14.19	218.72	2,769.2	-59.2	-35.7	66.9	2.22	1.64	6.20	
2,800.0	14.64	219.69	2,793.4	-64.0	-39.6	73.1	2.04	1.80	3.88	
2,825.0	15.16	220.61	2,817.6	-68.9	-43.7	79.5	2.29	2.08	3.68	
2,850.0	15.74	221.20	2,841.7	-73.9	-48.1	86.1	2.40	2.32	2.36	
2,875.0	16.41	221.66	2,865.7	-79.1	-52.7	93.1	2.73	2.68	1.84	
2,900.0	17.26	222.30	2,889.6	-84.5	-57.5	100.3	3.48	3.40	2.56	
2,925.0	17.89	222.65	2,913.5	-90.1	-62.6	107.8	2.56	2.52	1.40	
2,950.0	18.37	222.84	2,937.2	-95.8	-67.9	115.6	1.93	1.92	0.76	
2,975.0	18.91	222.78	2,960.9	-101.7	-73.3	123.6	2.16	2.16	-0.24	
3,000.0	19.63	223.07	2,984.5	-107.7	-79.0	131.8	2.91	2.88	1.16	
3,025.0	20.32	223.42	3,008.0	-113.9	-84.8	140.3	2.80	2.76	1.40	
3,050.0	21.12	223.24	3,031.4	-120.4	-90.9	149.2	3.21	3.20	-0.72	
3,075.0	21.82	223.54	3,054.7	-127.0	-97.2	158.3	2.83	2.80	1.20	
3,100.0	22.47	223.47	3,077.8	-133.8	-103.6	167.7	2.60	2.60	-0.28	
3,125.0	23.02	223.63	3,100.9	-140.8	-110.3	177.4	2.21	2.20	0.64	
3,150.0	23.70	223.59	3,123.8	-148.0	-117.1	187.3	2.72	2.72	-0.16	
3,175.0	24.37	223.34	3,146.7	-155.4	-124.1	197.5	2.71	2.68	-1.00	
3,200.0	25.00	223.32	3,169.4	-163.0	-131.3	207.9	2.52	2.52	-0.08	
3,225.0	25.75	223.42	3,192.0	-170.8	-138.7	218.6	3.00	3.00	0.40	
3,250.0	26.62	223.25	3,214.4	-178.8	-146.2	229.7	3.49	3.48	-0.68	
3,275.0	27.56	223.25	3,236.6	-187.1	-154.0	241.0	3.76	3.76	0.00	
3,300.0	28.53	223.08	3,258.7	-195.7	-162.1	252.8	3.89	3.88	-0.68	
3,325.0	29.28	223.07	3,280.6	-204.5	-170.3	264.8	3.00	3.00	-0.04	
3,350.0	30.08	223.18	3,302.3	-213.5	-178.8	277.2	3.21	3.20	0.44	
3,375.0	31.15	223.39	3,323.8	-222.8	-187.5	289.9	4.30	4.28	0.84	
3,400.0	31.93	223.37	3,345.1	-232.3	-196.5	303.0	3.12	3.12	-0.08	

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-131
Project:	SHAH	TVD Reference:	RTKB @ 483.7ft
Site:	SY-131	MD Reference:	RTKB @ 483.7ft
Well:	SY-131	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	SY-131	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
3,425.0	32.57	223.43	3,366.3	-242.0	-205.7	316.3	2.56	2.56	0.24
3,450.0	33.19	223.60	3,387.3	-251.9	-215.0	329.9	2.51	2.48	0.68
3,475.0	33.71	223.83	3,408.1	-261.8	-224.5	343.7	2.14	2.08	0.92
3,500.0	33.93	223.97	3,428.9	-271.8	-234.2	357.6	0.93	0.88	0.56
3,525.0	34.33	223.89	3,449.6	-281.9	-243.9	371.6	1.61	1.60	-0.32
3,550.0	34.97	223.92	3,470.2	-292.2	-253.8	385.8	2.56	2.56	0.12
3,575.0	35.09	223.80	3,490.6	-302.5	-263.7	400.2	0.55	0.48	-0.48
3,600.0	35.07	223.71	3,511.1	-312.9	-273.7	414.5	0.22	-0.08	-0.36
3,625.0	35.08	223.79	3,531.6	-323.3	-283.6	428.9	0.19	0.04	0.32
3,650.0	35.04	223.90	3,552.0	-333.6	-293.5	443.2	0.30	-0.16	0.44
3,675.0	35.04	223.97	3,572.5	-344.0	-303.5	457.6	0.16	0.00	0.28
3,700.0	35.02	223.92	3,593.0	-354.3	-313.5	471.9	0.14	-0.08	-0.20
3,725.0	35.43	223.69	3,613.4	-364.7	-323.4	486.3	1.72	1.64	-0.92
3,750.0	36.35	223.57	3,633.6	-375.3	-333.6	501.0	3.69	3.68	-0.48
3,775.0	37.27	223.39	3,653.6	-386.2	-343.9	516.0	3.71	3.68	-0.72
3,800.0	38.02	223.24	3,673.4	-397.3	-354.3	531.2	3.02	3.00	-0.60
3,825.0	38.89	222.91	3,693.0	-408.7	-365.0	546.8	3.58	3.48	-1.32
3,850.0	39.78	222.97	3,712.4	-420.3	-375.7	562.6	3.56	3.56	0.24
3,875.0	40.52	223.00	3,731.5	-432.1	-386.7	578.7	2.96	2.96	0.12
3,900.0	40.99	222.86	3,750.4	-444.0	-397.9	595.0	1.92	1.88	-0.56
3,925.0	41.18	222.49	3,769.2	-456.1	-409.0	611.4	1.23	0.76	-1.48
3,950.0	41.96	222.25	3,788.0	-468.3	-420.2	628.0	3.18	3.12	-0.96
3,975.0	42.81	222.16	3,806.4	-480.8	-431.5	644.8	3.41	3.40	-0.36
4,000.0	43.27	222.25	3,824.7	-493.5	-443.0	661.8	1.86	1.84	0.36
4,025.0	44.45	222.32	3,842.7	-506.3	-454.6	679.1	4.72	4.72	0.28
4,050.0	45.57	222.57	3,860.4	-519.3	-466.5	696.8	4.54	4.48	1.00
4,066.0	46.08	222.56	3,871.5	-527.8	-474.3	708.2	3.20	3.20	-0.08
9-5/8" PRODUCTION CASING									
4,075.0	46.37	222.55	3,877.8	-532.6	-478.7	714.7	3.20	3.20	-0.08
4,100.0	47.82	222.50	3,894.8	-546.1	-491.1	733.0	5.80	5.80	-0.20
4,125.0	49.91	222.34	3,911.2	-560.0	-503.8	751.8	8.37	8.36	-0.64
4,150.0	52.09	222.57	3,927.0	-574.3	-516.9	771.2	8.75	8.72	0.92
4,175.0	52.84	222.44	3,942.2	-588.9	-530.3	791.0	3.03	3.00	-0.52
4,200.0	53.52	222.39	3,957.2	-603.7	-543.8	811.0	2.72	2.72	-0.20
4,225.0	55.06	222.68	3,971.8	-618.6	-557.5	831.3	6.23	6.16	1.16
4,250.0	57.17	222.99	3,985.7	-633.9	-571.6	852.0	8.50	8.44	1.24
4,275.0	58.81	223.40	3,999.0	-649.3	-586.1	873.2	6.71	6.56	1.64
4,300.0	60.23	223.15	4,011.6	-665.0	-600.9	894.7	5.75	5.68	-1.00
4,325.0	61.89	222.95	4,023.7	-681.0	-615.8	916.6	6.68	6.64	-0.80
4,350.0	63.30	222.63	4,035.2	-697.3	-630.9	938.7	5.75	5.64	-1.28
4,375.0	64.62	222.18	4,046.2	-713.9	-646.1	961.2	5.52	5.28	-1.80
4,400.0	65.67	221.74	4,056.7	-730.7	-661.2	983.8	4.49	4.20	-1.76
4,425.0	67.43	221.79	4,066.7	-747.8	-676.5	1,006.7	7.04	7.04	0.20

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-131
Project:	SHAH	TVD Reference:	RTKB @ 483.7ft
Site:	SY-131	MD Reference:	RTKB @ 483.7ft
Well:	SY-131	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	SY-131	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
4,450.0	69.50	221.73	4,075.8	-765.2	-692.0	1,029.9	8.28	8.28	-0.24
4,475.0	70.65	221.30	4,084.4	-782.8	-707.6	1,053.3	4.88	4.60	-1.72
4,500.0	72.29	220.38	4,092.3	-800.7	-723.1	1,077.0	7.43	6.56	-3.68
4,525.0	73.49	220.09	4,099.7	-819.0	-738.5	1,100.8	4.93	4.80	-1.16
4,550.0	73.56	219.68	4,106.8	-837.3	-753.9	1,124.6	1.60	0.28	-1.64
4,575.0	73.57	219.52	4,113.8	-855.8	-769.1	1,148.5	0.62	0.04	-0.64
4,600.0	73.69	219.39	4,120.9	-874.3	-784.4	1,172.3	0.69	0.48	-0.52
4,625.0	75.07	219.74	4,127.6	-892.9	-799.7	1,196.3	5.68	5.52	1.40
4,650.0	76.32	220.53	4,133.8	-911.4	-815.3	1,220.4	5.86	5.00	3.16
4,675.0	77.75	221.46	4,139.4	-929.8	-831.3	1,244.7	6.77	5.72	3.72
4,700.0	79.81	222.49	4,144.3	-948.0	-847.7	1,269.2	9.18	8.24	4.12
4,725.0	81.25	223.66	4,148.4	-966.1	-864.6	1,293.8	7.38	5.76	4.68
4,750.0	81.81	224.26	4,152.1	-983.8	-881.7	1,318.5	3.26	2.24	2.40
4,756.5	82.12	224.41	4,153.0	-988.5	-886.2	1,324.9	5.29	4.77	2.31
4,837.0	87.31	226.91	4,160.4	-1,044.5	-943.6	1,405.1	7.14	6.44	3.10
7" PRODUCTION LINER									
4,840.0	87.50	227.00	4,160.5	-1,046.5	-945.7	1,408.1	7.14	6.45	3.08
4,845.0	87.50	227.00	4,160.7	-1,049.9	-949.4	1,413.1	0.00	0.00	0.00
4,846.4	87.51	227.01	4,160.8	-1,050.9	-950.4	1,414.5	1.01	0.71	0.71
4,882.9	87.73	227.30	4,162.3	-1,075.7	-977.2	1,450.9	1.00	0.60	0.79
4,929.2	87.73	227.30	4,164.1	-1,107.0	-1,011.1	1,497.1	0.00	0.00	0.00
SY-131 LP									
4,946.4	87.73	227.30	4,164.8	-1,118.7	-1,023.8	1,514.3	0.00	0.00	0.00
5,046.4	87.73	227.30	4,168.8	-1,186.4	-1,097.2	1,614.2	0.00	0.00	0.00
5,146.4	87.73	227.30	4,172.7	-1,254.2	-1,170.6	1,714.1	0.00	0.00	0.00
5,246.4	87.73	227.30	4,176.7	-1,322.0	-1,244.1	1,813.9	0.00	0.00	0.00
5,346.4	87.73	227.30	4,180.7	-1,389.7	-1,317.5	1,913.8	0.00	0.00	0.00
5,392.6	87.73	227.30	4,182.5	-1,421.1	-1,351.5	1,960.0	0.00	0.00	0.00
5,446.4	87.73	227.30	4,184.6	-1,457.5	-1,390.9	2,013.7	0.00	0.00	0.00
5,546.4	87.73	227.30	4,188.6	-1,525.3	-1,464.4	2,113.6	0.00	0.00	0.00
5,646.4	87.73	227.30	4,192.5	-1,593.0	-1,537.8	2,213.4	0.00	0.00	0.00
5,746.4	87.73	227.30	4,196.5	-1,660.8	-1,611.2	2,313.3	0.00	0.00	0.00
5,846.4	87.73	227.30	4,200.5	-1,728.5	-1,684.7	2,413.2	0.00	0.00	0.00
5,946.4	87.73	227.30	4,204.4	-1,796.3	-1,758.1	2,513.0	0.00	0.00	0.00
6,046.4	87.73	227.30	4,208.4	-1,864.1	-1,831.5	2,612.9	0.00	0.00	0.00
6,114.9	87.73	227.30	4,211.1	-1,910.5	-1,881.8	2,681.3	0.00	0.00	0.00
6,123.8	87.82	227.31	4,211.5	-1,916.5	-1,888.4	2,690.2	0.99	0.98	0.13
SY-131 MP 21 Oct 2018									
6,146.4	88.04	227.34	4,212.3	-1,931.8	-1,905.0	2,712.8	0.99	0.98	0.13
6,246.4	89.04	227.44	4,214.8	-1,999.5	-1,978.6	2,812.7	1.00	1.00	0.10
6,329.5	89.86	227.52	4,215.6	-2,055.7	-2,039.8	2,895.8	0.99	0.99	0.10
6,346.4	89.86	227.52	4,215.7	-2,067.1	-2,052.3	2,912.6	0.00	0.00	0.00
6,446.4	89.86	227.52	4,215.9	-2,134.6	-2,126.0	3,012.6	0.00	0.00	0.00

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-131
Project:	SHAH	TVD Reference:	RTKB @ 483.7ft
Site:	SY-131	MD Reference:	RTKB @ 483.7ft
Well:	SY-131	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	SY-131	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
6,546.4	89.86	227.52	4,216.1	-2,202.1	-2,199.8	3,112.5	0.00	0.00	0.00
6,646.4	89.86	227.52	4,216.4	-2,269.7	-2,273.5	3,212.4	0.00	0.00	0.00
6,746.4	89.86	227.52	4,216.6	-2,337.2	-2,347.3	3,312.4	0.00	0.00	0.00
6,846.4	89.86	227.52	4,216.9	-2,404.7	-2,421.0	3,412.3	0.00	0.00	0.00
6,946.4	89.86	227.52	4,217.1	-2,472.3	-2,494.8	3,512.2	0.00	0.00	0.00
7,046.4	89.86	227.52	4,217.4	-2,539.8	-2,568.5	3,612.2	0.00	0.00	0.00
7,146.4	89.86	227.52	4,217.6	-2,607.3	-2,642.3	3,712.1	0.00	0.00	0.00
7,246.4	89.86	227.52	4,217.9	-2,674.9	-2,716.0	3,812.1	0.00	0.00	0.00
7,324.9	89.86	227.52	4,218.0	-2,727.9	-2,774.0	3,890.6	0.00	0.00	0.00
SY-131 TD									

Design Targets									
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (ft)	+N/-S (ft)	+E/-W (ft)	Northing (m)	Easting (m)	Latitude	Longitude
SY-131 LP - actual wellpath misses target center by 18.2ft at 4929.2ft MD (4164.1 TVD, -1107.0 N, -1011.1 E) - Point	0.00	0.00	4,178.7	-1,114.7	-1,003.2	2,535,360.00	787,515.00	22° 54' 12.496 N	53° 48' 9.659 E
SY-131 TD - actual wellpath misses target center by 11.0ft at 7324.9ft MD (4218.0 TVD, -2727.9 N, -2774.0 E) - Point	0.00	0.00	4,213.7	-2,736.1	-2,779.8	2,534,865.80	786,973.50	22° 53' 56.777 N	53° 47' 50.345 E

Casing Points					
Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)	
190.0	190.0	13-3/8" CONDUCTOR	13.375	20.000	
4,066.0	3,871.5	9-5/8" PRODUCTION CASING	9.625	12.250	
4,837.0	4,160.4	7" PRODUCTION LINER	7.000	8.500	

Checked By: _____ Approved By: _____ Date: _____